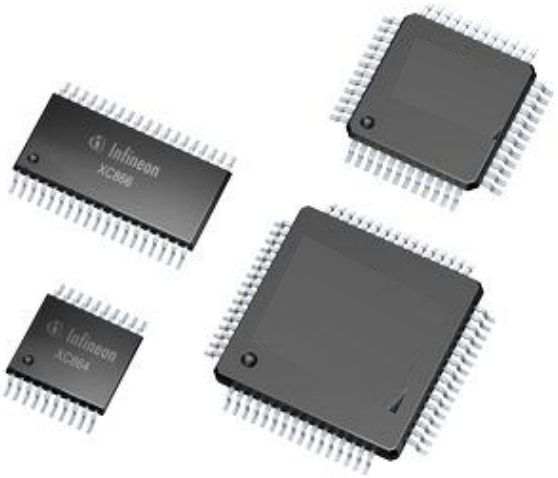
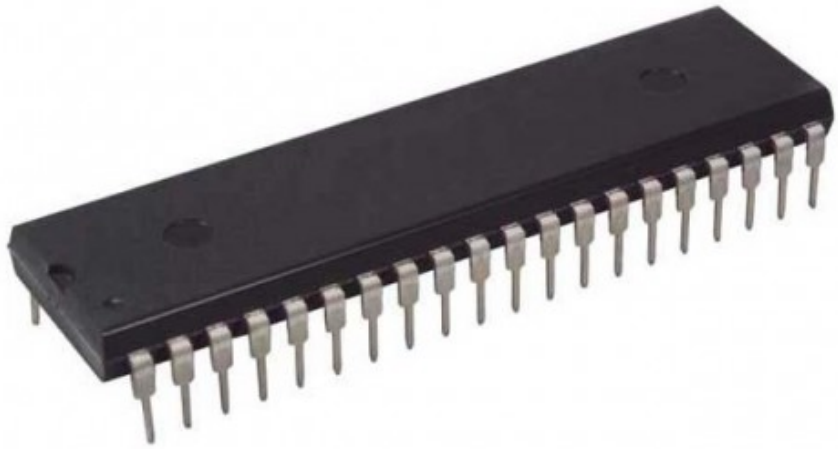
8051 Microcontroller





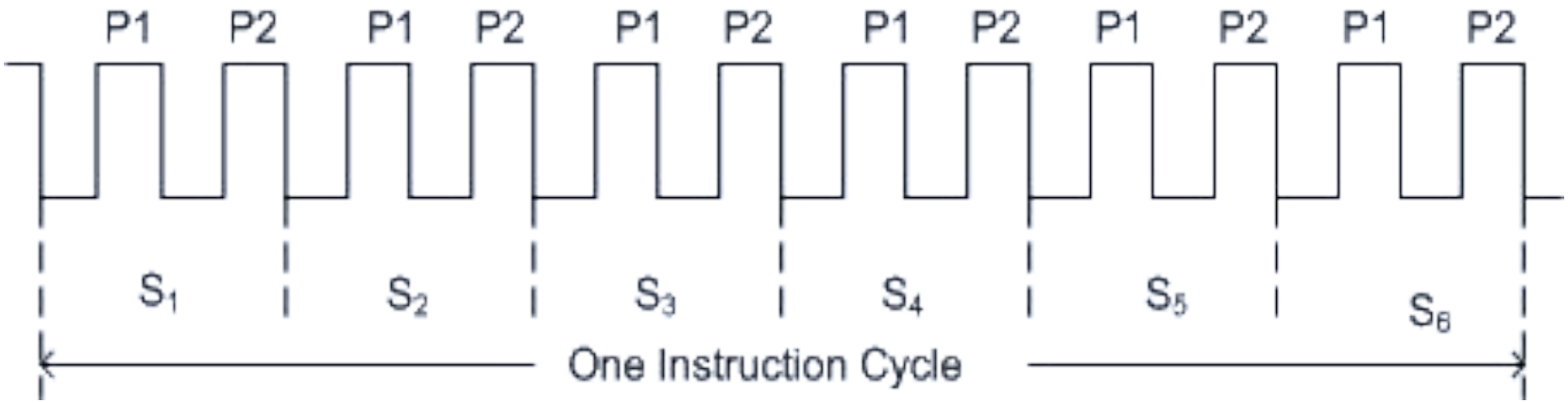
CPU timing And Machine cycle

- Oscillator Formed by crystal, capacitor and on-chip inverter generates a pulse train at frequency of crystal

- Pulse time: clock frequency established smallest interval of time within uc.

- State : basic time interval for discrete operation of uc such as fetching an opcode, decoding an opcode, executing an opcode

- Machine cycle: smallest interval of time to accomplish any simple instruction or part of complex instruction



1 MC= 6 State

1 State= 2 oscillator pulses

1MC= 12 clock period

Time required to execute instruction = $\frac{C \times 12}{\text{Crystal frequency}}$

Instruction	Machine cycle
ADD A, R0	1
MOV A, #23H	2
LJMP	3

XTAL = 11.0592 MHz

11.0592 MHz / 12 = 921.6 kHz;

machine cycle = 1 / 921.6 kHz = 1.085 µs